

VALENTIN PATAUD

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French (Driving licence B/EU)

PROFIL

Industrial Digitalization Engineer with 5 years of experience. Specialized in leading digital transformation projects in Gxp environment. Delivering measurable improvements in production performance and quality. Passionate about mechatronics, I also design and develop electronic and robotic systems through personal engineering projects.

EXPERTISE

Technical Skills

- Industrial Digitalization: MES, Digital Batch Records (DHR), Industry 4.0, process optimization
- Software Development: Java, JavaScript, C++, C#, SQL, Excel
- Low-Code Development & System Integration: Tulip, Node-RED, REST APIs, SAP integration
- Industrial Computing, IoT, Electronics, Electrical Systems (B1V electrical authorization)
- Data & Algorithms: Machine Learning, optimization, simulation, data analysis
- System Design: software architecture, PCB design, embedded systems development
- Validation & Compliance: GAMP 5, ISO 13485, 21 CFR Part 11

Professional Skills

- Technical project management & deliverables
- Business analysis & requirements gathering
- Continuous improvement & process optimization
- Cross-functional collaboration (Production, Quality, R&D, IT)
- Technical communication & knowledge transfer
- Experience working in international environments

LANGUES

- French: C2 – Native
- English: C1 – Fluent
- Italian: B1 – Professional working proficiency

FORMATIONS

MSC. IN COMPUTER SCIENCE, CONTROL AUTOMATION, ROBOTICS AND IT NETWORKS
Polytech Nancy - Engie Solutions Apprentice
2021-2024 With honors

BSC. IN AUTOMATION, ELECTRICITY AND COMPUTING SCIENCE
IUT de Nancy-Brabois
2019-2021 With honors

BACCALAURÉAT STI2D ITEC
2016-2019 With honors

INTERESTS

- Outdoor activities: cycling, trekking & winter sports
- Photography
- Design, development, and repair of electronic, mechatronic, and embedded systems (DIY projects)
- International culture

PROFESSIONAL EXPERIENCE

BIOMÉRIEUX ITALIA – FLORENCE, ITALIE
Digitalization Specialist (VIE) | Oct. 2024 – Oct. 2026

Digital transformation of manufacturing processes in regulated environments (GAMP 5, ISO 13485, 21 CFR Part 11).

- Design, development, and deployement of 15 applications (Tulip / Node-RED) to digitize DHR and industrial processes of 2 medical devices, generating 0.6 FTE (1,100 hours/year) annual savings across Production, Quality Assurance, and Engineering Support.
- Design and implementation of a Manufacturing Execution System (MES) integrated with SAP, for production order management, serial number traceability, production time tracking, and the elimination of paper-based records (116 hours/year saved).
- Leading project activities including business analysis, computerized system validation (NPS), SAP integration, API development, multisite deployment, and industrial support.

ENGIE SOLUTIONS – NANCY, FRANCE
Project engineer in Electricity and Automation (Apprentice)
Sep. 2021 – Sep. 2024

Creation of software solutions supporting energy transition initiatives and the automation of engineering studies.

- Design and deployement of EV charging infrastructure, photovoltaic systems sizing and cost estimation tools used for commercial engineering.
- Developement of optimization algorithms (graph theory, combinatorial optimization, Machine Learning) for technical sizing and system design.
- Creation of a Smart Grid prototype for intelligent building energy management using Machine Learning-based behavioral simulation..
- Technical reference for EV charging infrastructure and photovoltaic solutions, supporting commercial teams and standardizing methodologies.
- International assignment at Engie Italy: developed a data centralization platform prototype using Python, PostgreSQL, and Power BI.

EPROLOR – NANCY, FRANCE
Software devloper (Internship) | Avr. 2021 – Juin 2021

- Developed CMMS using C# and SQL.
- Redesigned maintenance processes and digitized intervention tracking workflows.

REFERENCES

FULVIO DIARA
Industrial Support & Engineering team manager @ bioMérieux Italia
+390556449856
Fulvio.daria@biomerieux.com

DIDIER MEYER
Executive project director @ Engie Solutions
+33607294787
didier.meyer@engie.com

PERSONAL ENGINEERING PROJECTS

Creation of a 3-Axis Motion Control System for DSLR Cameras (3 years)

- Designed and built a fully custom motorized camera motion control system inspired by professional solutions Syrp Genie II and Magic Carpet
- Full lifecycle: SolidWorks CAD design, PCB, mechanical manufacturing, hardware integration, firmware development, and Android app creation.

Nixie Tube Clock (6 months)

- Full Design and manufacturing of a custom Nixie tube clock with PCB design, firmware, hardware integration. Handcrafted using wood and Damascus steel.